

Physics 201 - Lecture 10

① Test on Friday - on WebAssign, during regular class time.

② Assignment 3 due on Wednesday.

③ Plan for today:

(i) A3 - # 3, # 4, # 5

(ii) Uniform Circular Motion.

(iii) A3 - # 11, # 12

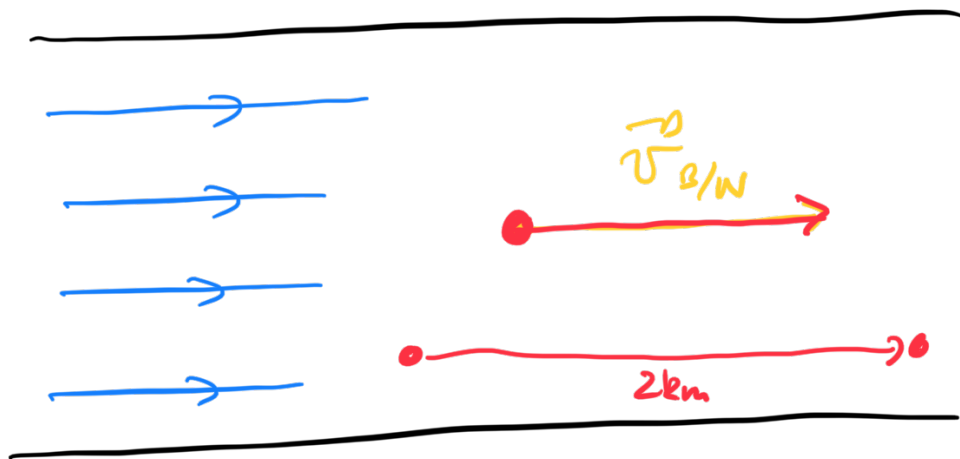
#3. A boat can be rowed at
... in ... still water.

$$7.5 \text{ km/hr in } \dots$$

$$|\vec{v}_{B/W}| = 7.5 \text{ km/hr}$$

Reference frames: ① ② ③
 BOAT, WATER, SHORE
 (B) (W) (S)

DIAGRAM !!



$$\vec{v}_{W/S} = 2.7 \hat{i}$$

$$\vec{v}_{B/S} = \vec{v}_{B/W} + \vec{v}_{W/S}$$

$$= 7.5 \hat{i} + 2.7 \hat{i}$$

$$\vec{v}_{B/S} = 10.2 \hat{i}$$

v/s

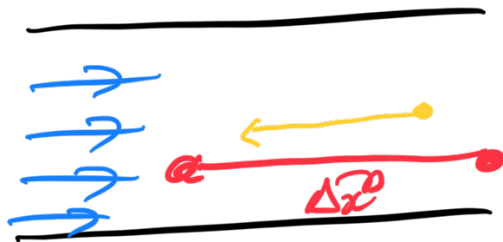
$$\Delta \vec{x}^0 = 2.0 \hat{i}$$

$$\Delta \vec{x}^0 = \vec{v}^0 \cdot t \Rightarrow t = \frac{\Delta \vec{x}^0}{\vec{v}^0_{B/s}}$$

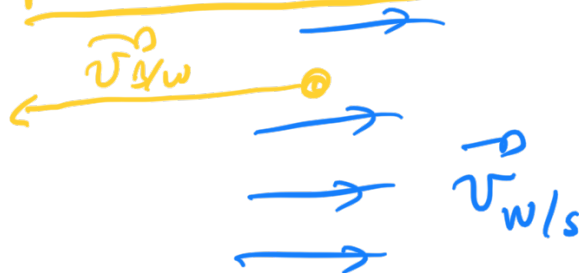
$$= \frac{2.0 \text{ km} \cancel{\text{h}}}{10.2 \cancel{\text{km}} / \cancel{\text{hr}}}$$

$$= 0.1961 \text{ hr}$$

$$= 11.8 \text{ min}$$



b)
$$\vec{v}_{B/w} = -7.5 \hat{i}$$



$$\vec{v}_{B/s} = \vec{v}_{B/w} + \vec{v}_{w/s}$$

$$= -7.5 \hat{i} + 2.7 \hat{i}$$

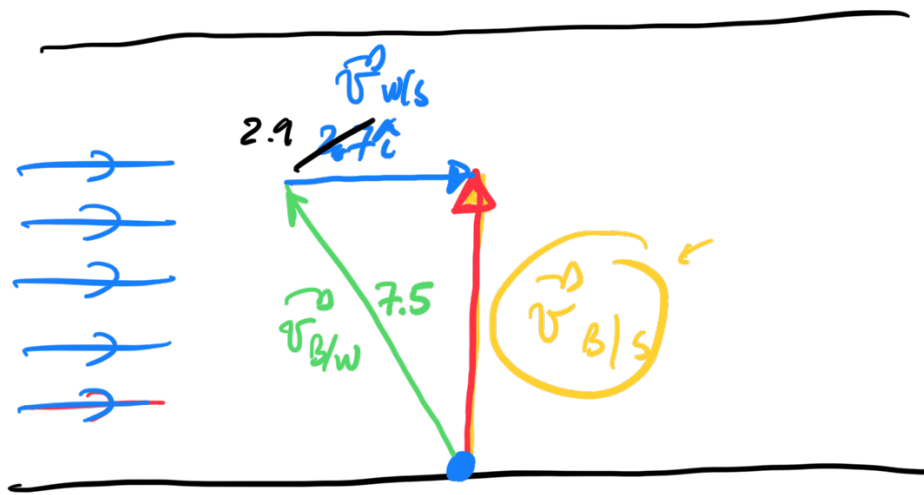
$$= -4.8 \hat{i}$$

$$\Delta \vec{x}^0 = \underline{-2.0 \hat{i}}$$

(rows back to original position)

$$t = \frac{\Delta \vec{x}}{\vec{v}} = \frac{-2.0 \hat{i}}{-4.8 \hat{i}} = 0.4167 \text{ hr} = 25 \text{ min.}$$

c)



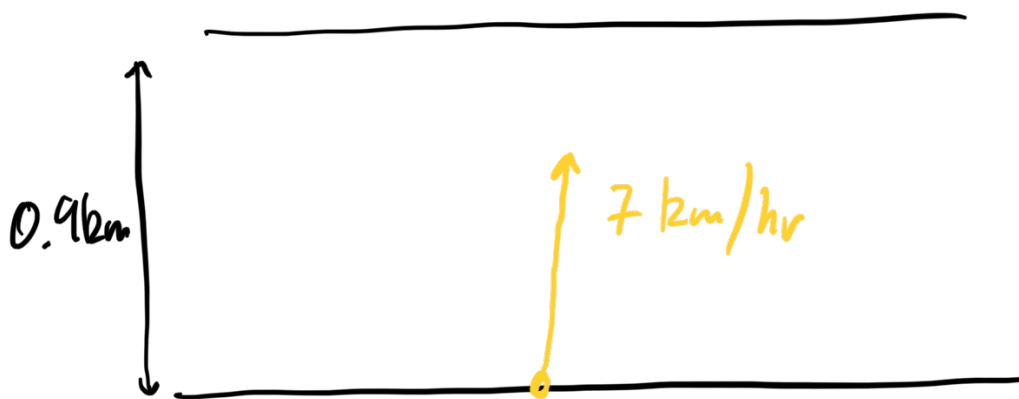
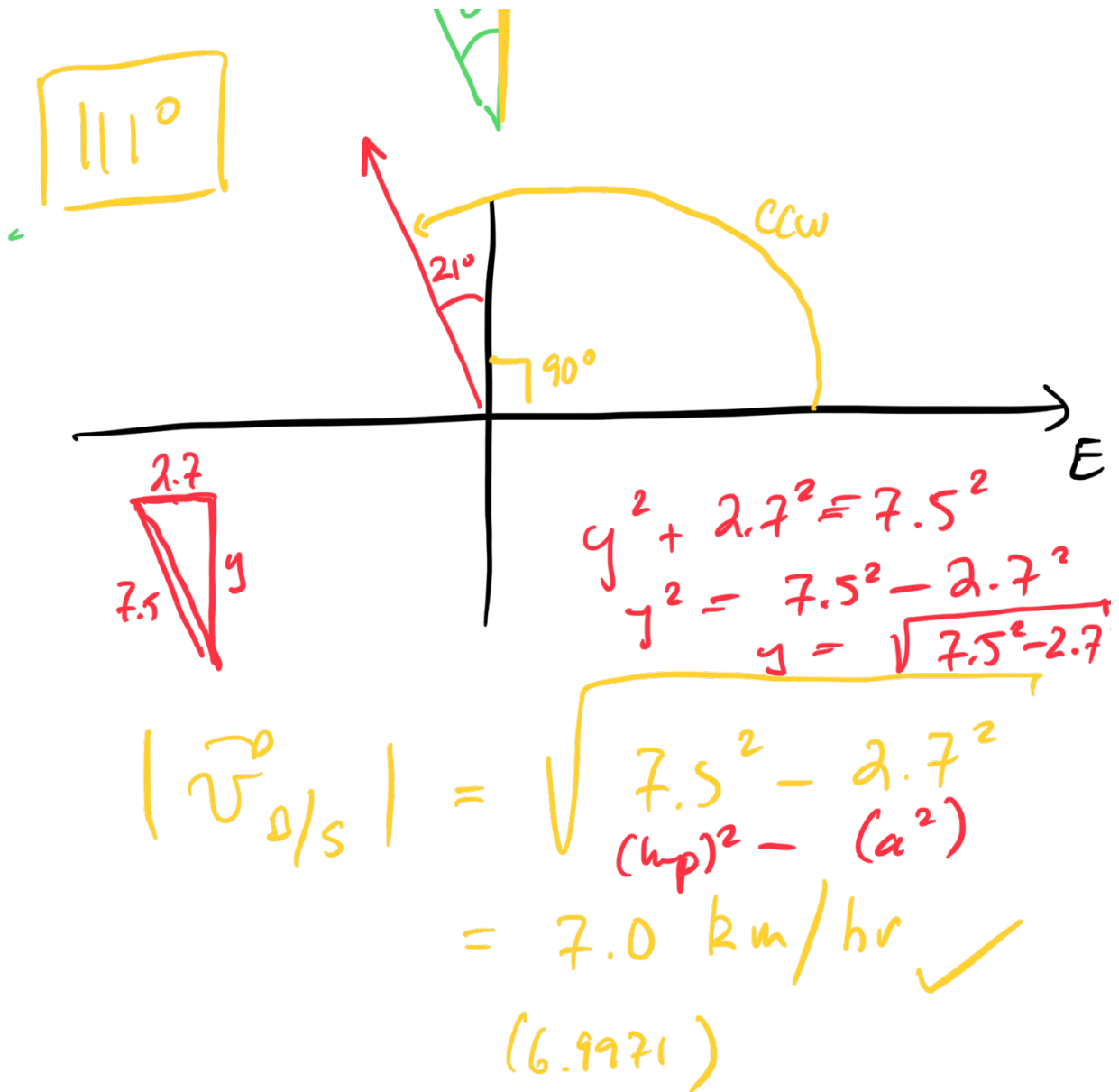
resultant $\vec{v}_{B/s} = \vec{v}_{B/w} + \vec{v}_{w/s}$ $(2.7 \hat{i})$

PICTURE



$$\sin \theta = \frac{2.7}{7.5} \quad \left(\frac{op}{hy} \right)$$

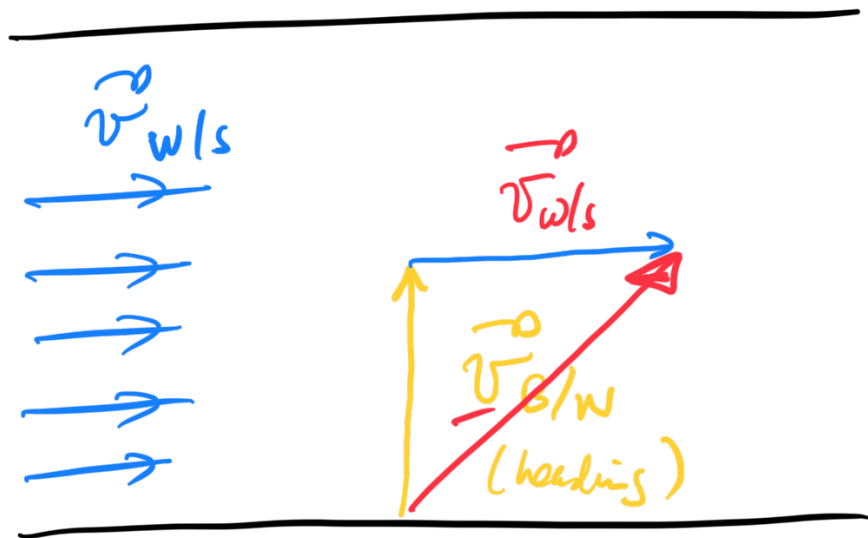
$$\theta = 21^\circ$$



$$t = \frac{d}{v} = \frac{0.9 \text{ km}}{7 \text{ km/hr}}$$

$$\begin{aligned}
 & + 2 \text{ m/hr} \\
 & = 0.1286 \text{ hours} \\
 & = 7.72 \text{ min}
 \end{aligned}$$

(e)

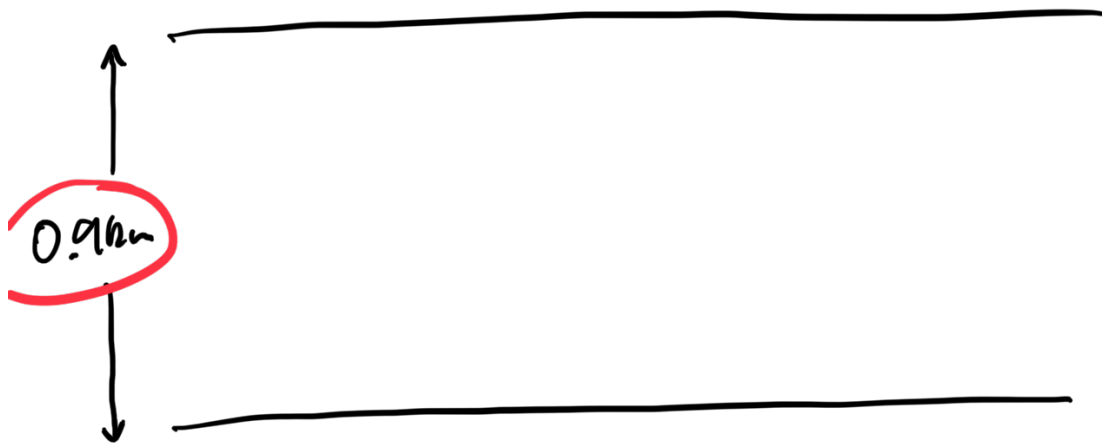


$$\begin{aligned}
 \vec{v}_{B/s} &= \vec{v}_{B/w} + \vec{v}_{w/s} \\
 &= 7.5 \hat{j} + 2.7 \hat{i}
 \end{aligned}$$



$$h = \sqrt{2.7^2 + 7.5^2}$$

$$|\vec{v}_{B/s}| = 7.9712 \text{ km/h}$$



$$t = \frac{0.9 \text{ km}}{7.5 \text{ km/hr}} \quad \leftarrow \Delta y \quad \leftarrow \textcircled{V_y}$$

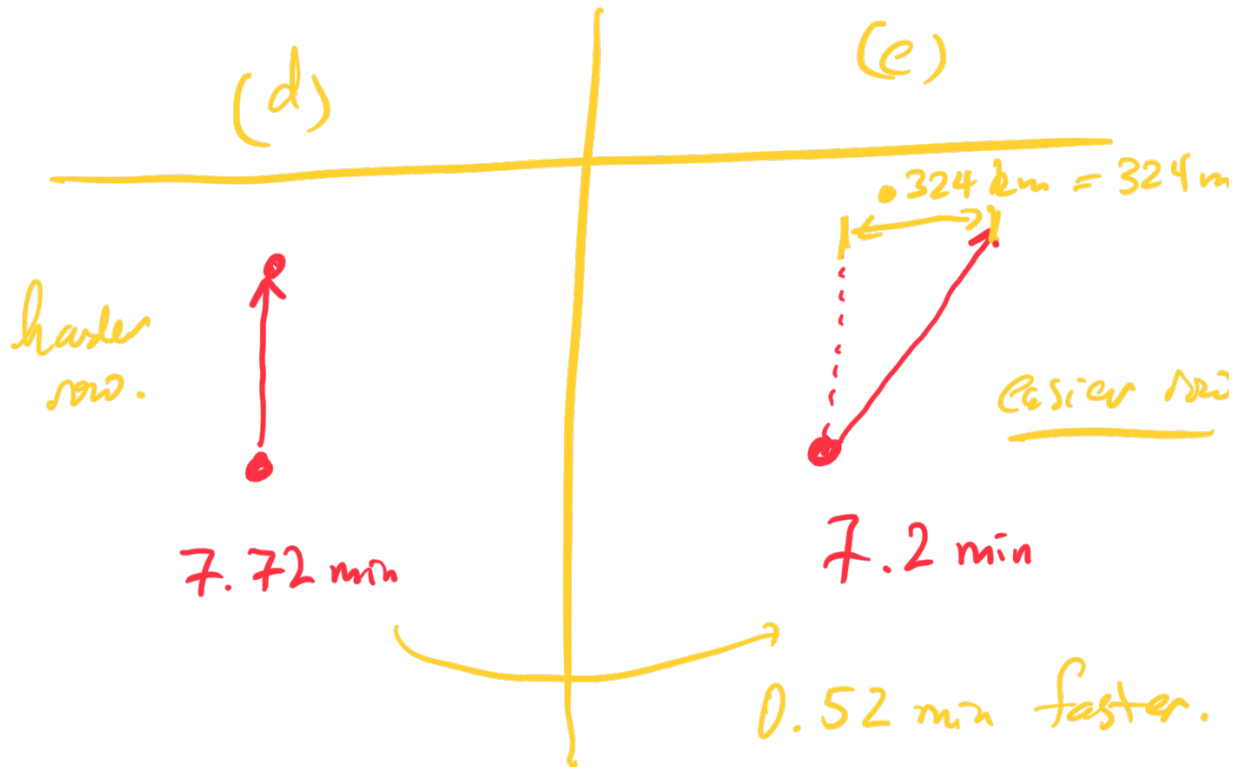
$$= 0.12 \text{ hours}$$

$$= 7.2 \text{ min.}$$

$$V_x = 2.7 \text{ km/hr}$$

$$\Delta x = 2.7 \frac{\text{km}}{\text{hr}} \times 0.12 \text{ hr}$$

$$= 0.324 \text{ km.}$$

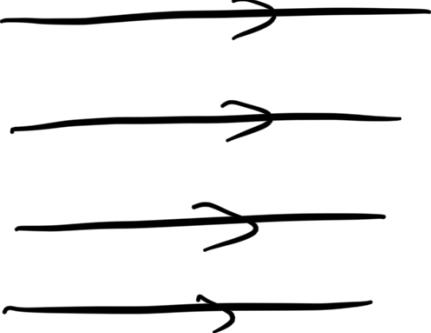


$$\frac{324 \text{ m}}{31.2 \text{ s}} = 10.38 \text{ m/s}$$

$$\sim \underline{\underline{20 \text{ mph.}}}$$

$$44.52 \text{ s} \times \frac{324}{400} = \underline{\underline{36.1 \text{ s}}}$$

#4. $\left| \vec{v}_{P/A} \right| = 190 \text{ km/hr}$

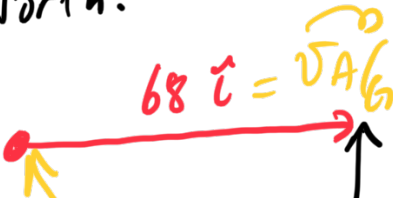
ω  $\vec{v}_{A/G} = 68 \hat{i}$

Ref: P, A, G
 ① ② ③

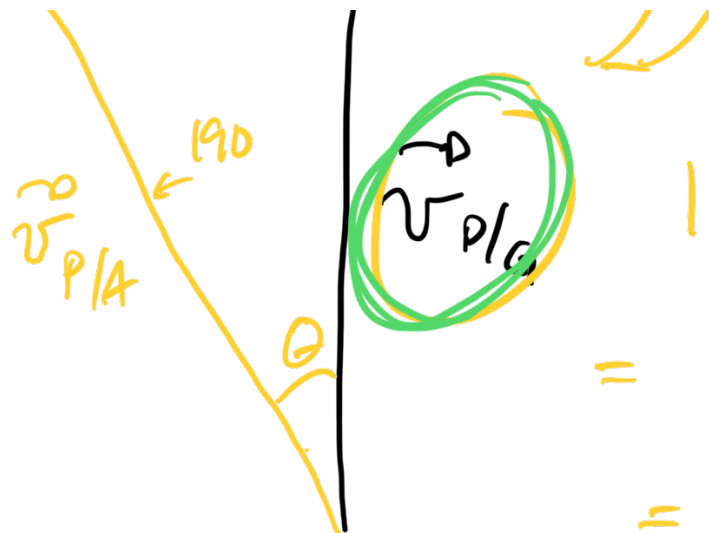
$\vec{v}_{P/G} = \vec{v}_{P/A} + \vec{v}_{A/G}$ (size.)

$\uparrow$
 North.

$68 \hat{i} = \vec{v}_{A/G}$ (68 $\hat{i}$)



4



$$|\vec{v}_{p/G}| = \sqrt{190^2 - 68^2} = 177.41 \text{ km/h}$$

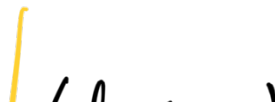
$$\sin \theta = \frac{68}{190} \Rightarrow \theta = 21^\circ$$

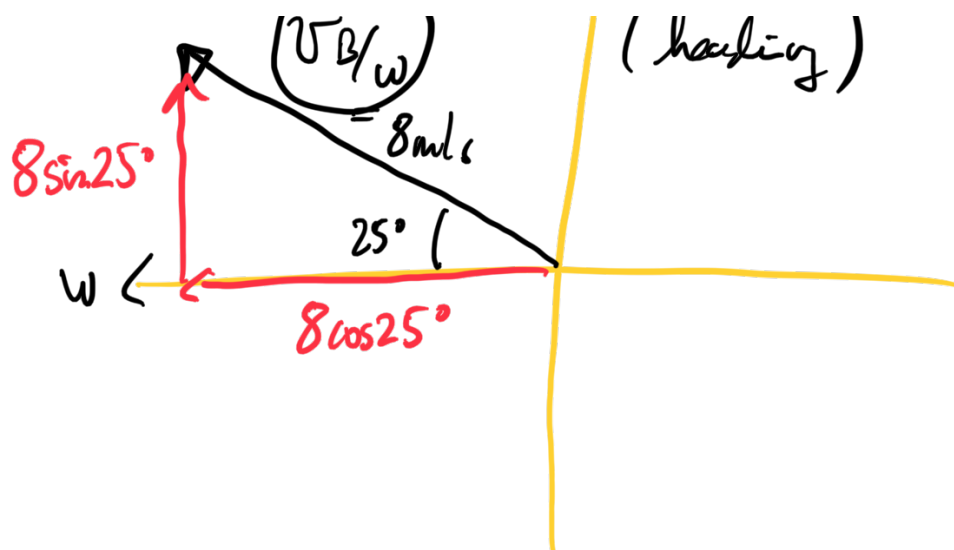
West of North

$$t = \frac{\Delta y}{|\vec{v}_{p/G}|} = \frac{400 \text{ km}}{177.41 \text{ km/h}} = 2.25 \text{ hours}$$

#5.

$$\vec{v}_{w/s} = 5 \hat{i}$$





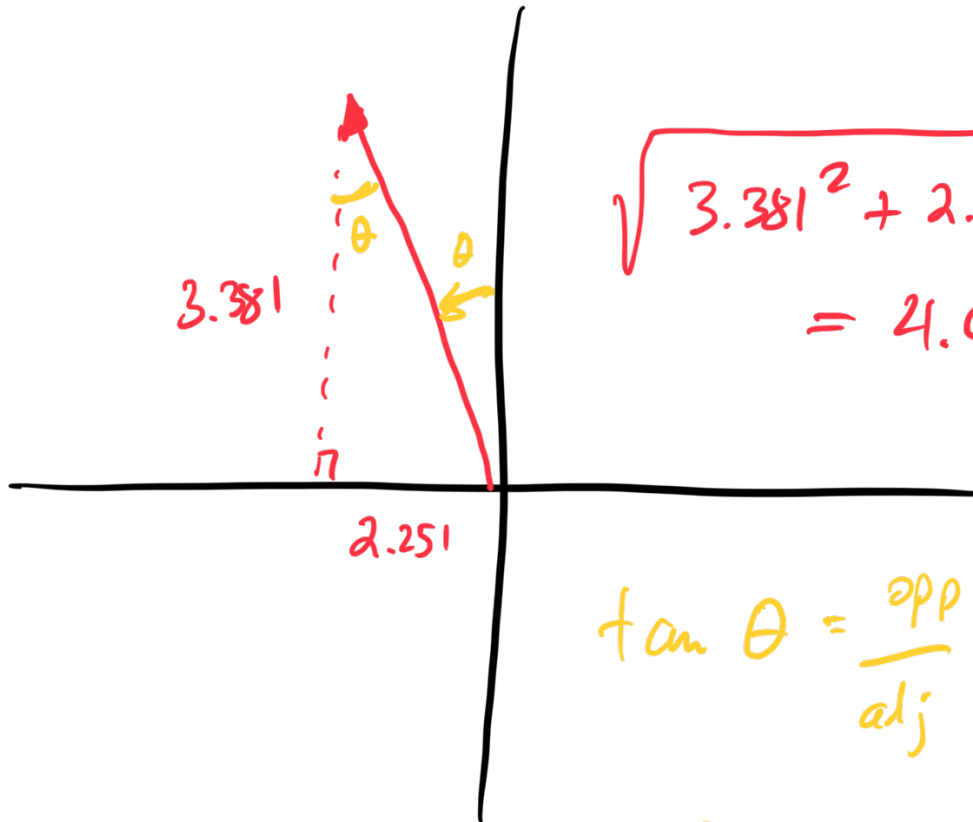
$$\vec{v}_{B/W} = -8 \cos 25^\circ \hat{i} + 8 \sin 25^\circ \hat{j}$$

$$\vec{v}_{B/W} = -7.251 \hat{i} + 3.381 \hat{j}$$

$$\vec{v}_{W/S} = 5 \hat{i}$$

$$\begin{aligned} \vec{v}_{B/S} &= \vec{v}_{B/W} + \vec{v}_{W/S} \\ &= -7.251 \hat{i} + 3.381 \hat{j} + 5 \hat{i} \end{aligned}$$

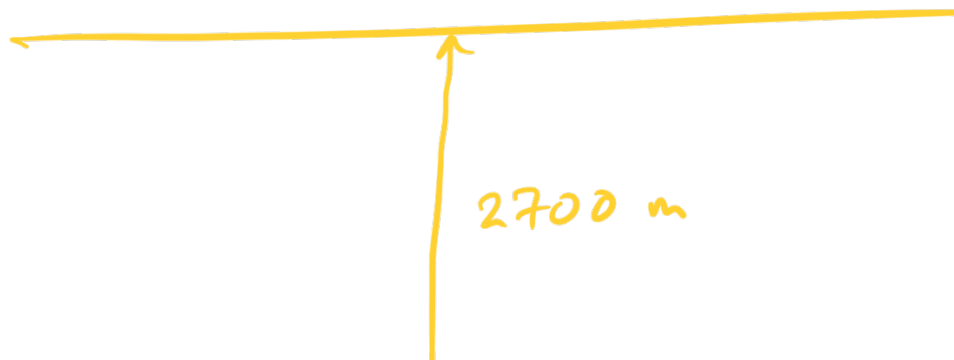
$$\vec{v}_{B/C} = -2.251 \hat{i} + 3.381 \hat{j} \quad \checkmark$$



$$\sqrt{3.381^2 + 2.251^2} = 4.06 \text{ m/s}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{2.251}{3.381}$$

$$\theta = 33.6^\circ \text{ W of K}$$



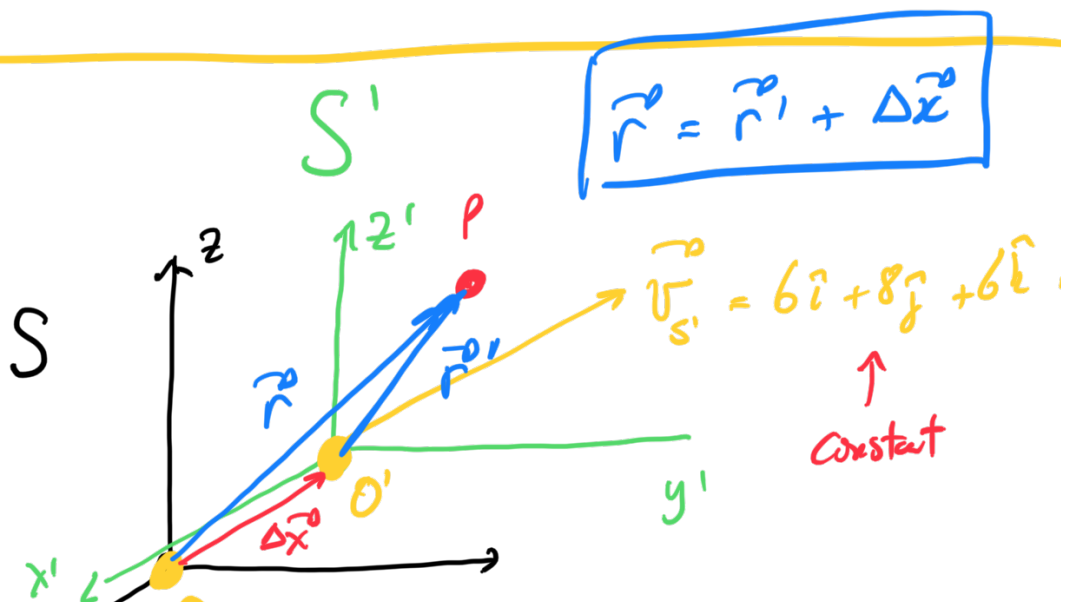
$$t = \frac{\Delta y}{v_y} = \frac{2700 \text{ m}}{3.381 \text{ m/s}}$$

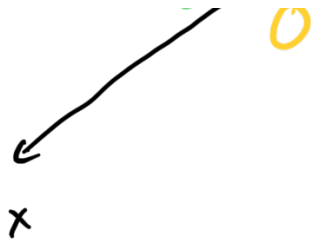
$$= 798.5 \text{ s}$$

$$= 13.3 \text{ min}$$

BYE!

#6.



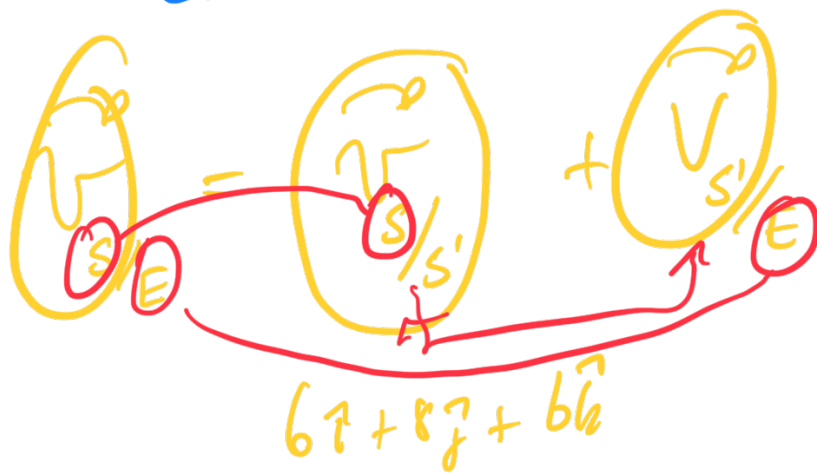


y

$$\Delta \vec{r} = \vec{v}_{s'} \cdot t + \frac{1}{2} \vec{a}_{s'} t^2$$

$$= (6\hat{i} + 8\hat{j} + 6\hat{k}) t$$

$$\Delta \vec{a} = 6t \hat{i} + 8t \hat{j} + 6t \hat{k}$$



$$\vec{v}_s = \vec{v}_{s'} + (6\hat{i} + 8\hat{j} + 6\hat{k})$$

$$\vec{a}_{s/E} = \vec{a}_{s/s'} + \vec{a}_{s'/E}$$

0

~ ~ ~

$$\dot{a}_s = a_{s'} + U$$